ReCell | Public controller Mathematical review supplement

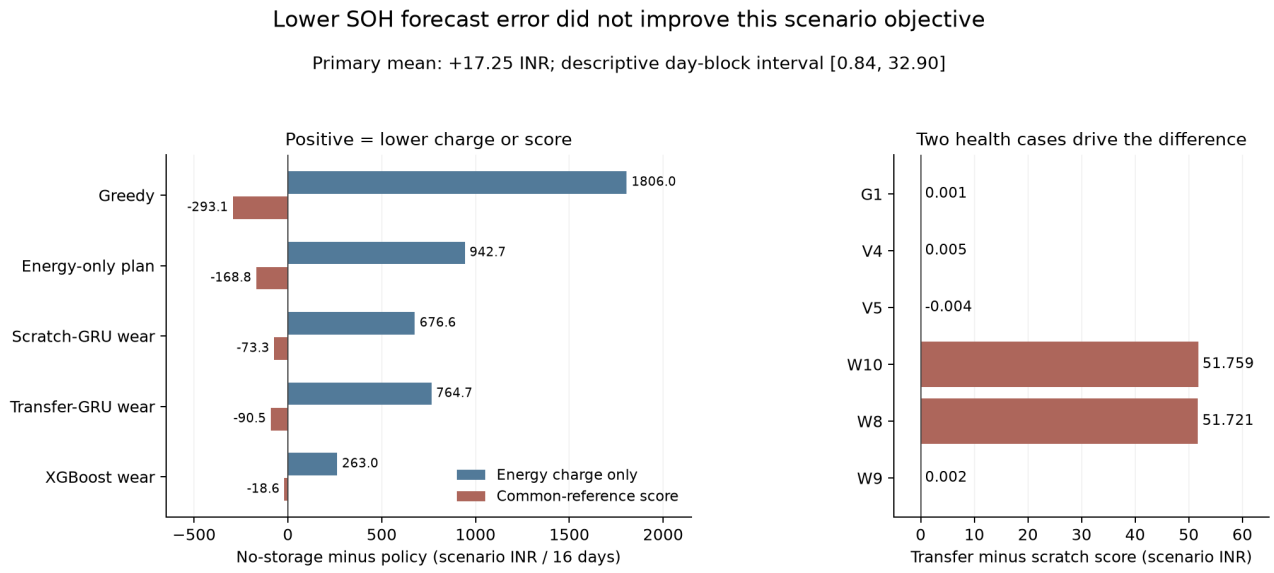
30 October 2026 | India C&I; research | Registered offline scenario

Review decision: advance data access and calibration for a validation pilot. The experiment does not support autonomous operation, an investment approval or guaranteed savings. The published research dashboard remains unchanged.

Transfer-based planning changes the 16-day common-reference score by **+INR17.25** versus scratch-based planning; higher is worse. Descriptive 95% day-block interval: [0.84, 32.90] INR.

Policy	Energy charge	Wear proxy	Inventory	Reference score
No battery	43,392.82	0.00	0.00	43,392.82
Greedy	41,586.79	1,962.68	136.42	43,685.89
Energy-only plan	42,450.09	1,003.95	107.61	43,561.65
Scratch-GRU wear	42,716.26	781.53	-31.67	43,466.11
Transfer-GRU wear	42,628.10	875.11	-19.85	43,483.36
XGBoost wear	43,129.87	314.60	-33.04	43,411.43

Mean scenario INR per 16 days: average three seeds within each health case, then six health cases equally. All policies within a case share physical capacity and limits. Amounts are not measured utility bills or observed physical wear.



One site, 16 days, six hypothetical battery-health cases, three model seeds. Hourly modeled PV and ideal interval recourse.
Reference score includes training-derived wear and terminal inventory. These are not measured bills, physical wear or field savings.

Lower grid energy charges do not automatically improve the complete objective. No battery has the lowest mean reference score here; demand/ToD charges, actual pack calibration and investment costs remain outside this scenario.

1. Data, attributes and prediction

Attribute	Use and boundary
Diagnostic discharge capacity Q (Ah)	Measured health target; Q_ref is the first accepted capacity of that life stage, not a future maximum.
Past SOH, elapsed stage days, cumulative Ah	Three shared GRU/XGBoost predictors. Only accepted history available at an origin enters the sequence.
Physical battery, life stage, hashes and protocol	Split/provenance keys. Histories do not cross life stages; a held-out cell's first life cannot train its second-life predictor.
Delhi academic-building demand (kW)	Measured active-power benchmark: 91 complete days, chronological 60/15/16-day split. Gross/net-meter topology remains unverified.
Regional hourly GHI and modeled PV (kW)	NASA historical resource; 40 kWp DC/32 kW AC horizontal scenario. Quarter-hour values retain 60-minute source resolution; not measured PV.
Temperature, resistance, duty and future metadata	Not predictors in the three-feature transfer comparison. Ambient air is not battery temperature. Future labels/exposure never fill past inputs.

```
SOH_j = Q_j / Q_ref
age_j = (diagnostic_end_j - diagnostic_end_1) / 86400
Ah = sum_k[0.5*(abs(I_k)+abs(I_(k+1)))*delta_t_k/3600]
x_j = [SOH_j, age_j, verified_Ah_j - verified_Ah_1]
```

The registered v3 amendment uses verified first-life prefixes and separate later measured capacity labels. The original full-history v2 gate remains failed. A future label can supervise a valid earlier input; it cannot create a missing input history.

Source training predicts the next checkpoint; the target model outputs the next two checkpoints. Transfer copies the GRU/representation, resets the head and optimizer, and fine-tunes on permitted target-training cells. Scaling uses target-training checkpoints only. Six physical test folds retain 87 origins and three seeds.

```
RMSE_b,s = sqrt(mean_origins[(predicted_SOH - measured_SOH)^2])
Reported RMSE = mean_cells(mean_seeds(RMSE_b,s)) * 100
```

10% context, next checkpoint	RMSE (SOH percentage points)
Scratch PGP-GRU	0.2972
Transfer PGP-GRU	0.2916
XGBoost	0.2534

Transfer's small average health-error improvement has negative cases (G1/W8). Nominal 80% intervals are not well calibrated. XGBoost has the lowest observed mean error. Six familiar cells do not establish population superiority.

2. Scheduling physics and information

The comparison replays one public demand window, 1-16 December 2014. Six later laboratory health states with three model seeds create 18 hypothetical cases, not 18 sites. The lab states are not attributed to the building in 2014.

Let L_t and S_t be interval mean load and modeled PV (kW); c_t and d_t are AC charging/discharging power. e_t is stored internal energy (kWh). eta is one-way efficiency. The 40 kWh/20 kW battery is generic, with constant health during replay.

```
dt = 0.25 h; eta = sqrt(0.90); E_nom = 40 kWh
H_origin = origin_SOH * Q_ref / 4.85 Ah
E_capacity = E_nom * H_origin
0.1*E_capacity <= e_t <= 0.9*E_capacity
e_initial = 0.5*E_capacity
e_(t+1) = e_t + eta*c_t*dt - d_t*dt/eta

surplus_t = max(S_t-L_t, 0); deficit_t = max(L_t-S_t, 0)
0 <= c_t <= min(20, surplus_t)
0 <= d_t <= min(20, deficit_t)
direct_PV_t = min(L_t,S_t)
grid_import_t = deficit_t - d_t
curtailed_PV_t = surplus_t - c_t; grid_export_t = 0
throughput = sum_t[(eta*c_t+d_t/eta)*dt]
EFC = throughput / (2*E_nom)
```

Charging is PV-only; discharge serves net load. Nonoverlapping surplus and deficit prohibit simultaneous charge/discharge. Residual PV is curtailed; grid imports are unlimited. Export permission, backup operation and physical curtailment capability have not been validated at a site.

Planning at midnight IST: forecast the next 96 quarters using completed past intervals. Demand is the frozen 50/50 previous-day/previous-week blend; PV repeats the previous day's modeled profile. No future realized load, resource or SOH label enters the plan.

```
Lhat(day,q) = 0.5*L(day-1,q) + 0.5*L(day-7,q)
Shat(day,q) = S(day-1,q)
```

Execution: ideal current-quarter feedback clamps requested power to realized mean surplus/deficit, power and energy bounds. Realized SOC carries between quarters and days without a hard daily reset. A zero-latency modeled observation feed and interval-mean feedback are not a verified live controller.

Unit example: charging at 10 kW for 15 minutes consumes 2.5 AC kWh and adds approximately 2.372 internal kWh at eta = sqrt(0.90). Energy is not kW multiplied by 15 without converting minutes to hours.

3. Wear, costs and the objective

Only four target gradient-training cells calibrate wear for each held-out case. Validation/test histories and future SOH labels never calibrate it. Let delta_Ah be accepted interval exposure, and Q_ref the cell's stage reference.

```
EFC_b,j = delta_Ah_b,j / (2*4.85 Ah)
loss_b,j = (SOH_(j-1)-SOH_j)*Q_ref_b/4.85 Ah
expected_EFC = median_b(median_j(EFC_b,j))
r_reference = max(0, median_b(median_j(loss_b,j/EFC_b,j)))
r_planner = max(0, (origin_SOH-predicted_median_SOH)
* Q_ref/4.85 Ah / expected_EFC)
```

The medians use only exposure increments above the audited 1e-6 Ah tolerance. Signed capacity recoveries remain in reference-loss statistics; zero-exposure intervals receive no invented rate. XGBoost supplies a point forecast, without fabricated quantiles. This mixed-aging throughput proxy is not physical wear truth; no calendar term is added.

```
price = 10 INR/kWh; replacement = 400000 INR; loss_budget = 0.20
Wear(r) = replacement * r * EFC / loss_budget
v = price*eta [INR/internal kWh]
J_plan = price*forecast_grid_kWh + Wear(r_planner)
- v*(e_end-e_start) + 1e-6*sum_t[(c_t^2+d_t^2)*dt]
J_reference = price*realized_grid_kWh + Wear(r_reference)
- v*(e_final-e_initial)
```

Terminal inventory prevents free use of initial energy and accounts for unequal final SOC. The credit is an assumed energy-service value, not a utility payment or full future wear/value model. The fixed quadratic planning preference is excluded from evaluation; its maximum INR0.0192/day is not a replay error bound.

Transfer minus scratch component	Scenario INR / 16 days
Energy charge	-88.16
Common reference wear	+93.58
Inventory adjustment	+11.82
Total reference score	+17.25

Evaluation-wear multiplier	Transfer minus scratch score (INR)
0	-76.34
0.5	-29.54
1	+17.25
2	+110.83

These fixed-action score sensitivities do not reoptimize policies. The ranking changes with an unvalidated coefficient. Totals are calculated before rounding. W10/W8 dominate the primary difference; four other case differences are below INR0.006. No result is a measured saving.

4. Evidence gates and review decision

Evidence	Current finding / next required evidence
Reference and transfer	539 frozen processed/result/dashboard paths unchanged. v3 comparison complete; v2 source defects remain. Keep all physical splits and exclusions.
Calendar application	Separate study complete, including 44 negative predictions. M50T calibration, probe mapping, loss decomposition and intended-use clarification remain open.
Paired site measurements	Measured load plus modeled PV only. Obtain actual paired load/PV/battery records, meter topology, units, timezone, quality flags and reuse terms.
Connection and bill	Obtain category, voltage, contract demand, applicable DISCOM amendments, ToD/demand/export rules and complete bills. Reconcile kVAh/PF where required.
Pack and BMS	Obtain chemistry/identity, quotation, measured usable capacity/efficiency, SOC/temperature channels, sensor map and operating constraints.
Forecasts and operation	Seal independent site split, forecast availability, delay, policy arms and success criteria before scoring. Verify timing and perform supervised operation.
Dashboard release	Hold candidate release until provenance, selection and field/claim gates pass. Current public simulation cannot approve deployment or establish lifetime extension.

Verification: the latest recorded software suite has 216 passing tests. Independent reviews replayed all 15,660 health predictions and reconstructed all 165,888 controller actions and 108 case scores. Forecast cutoffs, SOC continuity, wear coefficients and primary bootstrap were checked. Maximum planner residual is 1.95e-8; realized balance/SOC residual is 3.56e-15 kWh. The new integration audit checks saved artifact bytes and review links, not hardware.

Uncertainty: the primary controller 95% interval uses 10,000 paired circular four-day block resamples over 16 days, after averaging seeds then health cases. It is conditional on one short site window, modeled PV, fixed pack and wear assumptions. It is not cross-site or population evidence.

Sources and reproducibility

Demand: I-BLEND, DOI 10.6084/m9.figshare.6007637.v1 (CC0); modeled solar resource: NASA POWER, approximate point 28.55 N / 77.27 E. Detailed source/units audits: docs/india_public_data_audit.md and docs/india_pv_source_audit.md.

Registered controller protocol: configs/india_public_controller_v1.yaml. Sealed 29 October 2026, 17:15:12 UTC, before controller outcomes. Health results, load errors and PV totals were already known; no new prospective cohort is claimed.

Equations/results: docs/india_controller_review.md. Complete requirement review: docs/integration_gate_review.md. Executed runs, ledgers and independent verification: outputs/india_ci/india_public_controller_v1/.

Review provenance: outputs/integration/gate_review_20261030/. The original attributes/model, transfer and calendar PDFs are preserved. This supplement documents the separate quarter-hour controller; it does not rewrite the hourly historical optimizer.

Decision boundary: discuss a validation pilot for missing data and calibration. An actual site's acceptance criteria require agreement before evaluation. The full phases 14-20 goal remains incomplete.